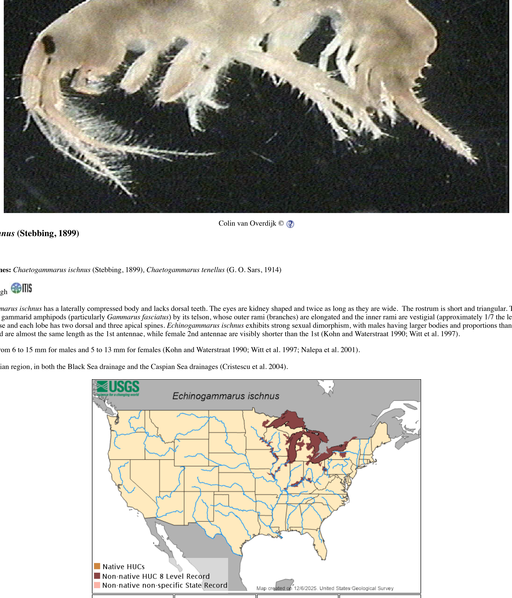


Echinogammarus ischnus (scud)
Crustaceans-Amphipods
Exotic

Collection Info
Point Map
Species Profile
Animated Map



Colin van Overdijk ©

Echinogammarus ischnus (Stebbing, 1899)

Common name: scud

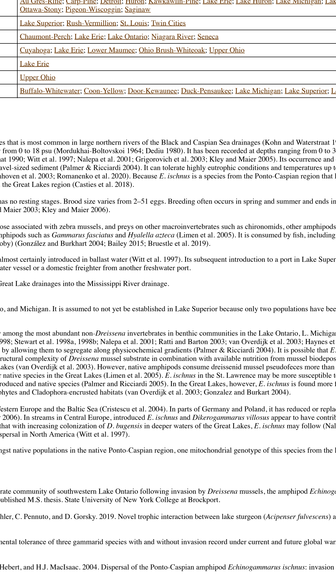
Synonyms and Other Names: *Chaetogammarus ischnus* (Stebbing, 1899), *Chaetogammarus tenellus* (G. O. Sars, 1914)

Taxonomy: available through 

Identification: *Echinogammarus ischnus* has a laterally compressed body and lacks dorsal teeth. The eyes are kidney shaped and twice as long as they are wide. The rostrum is short and triangular. This species is distinguishable from native gammarid amphipods (particularly *Gammarus fasciatus*) by its telson, whose outer rami (branches) are elongated and the inner rami are vestigial (approximately 1/7 the length of the outer pair). The telson is also cleft to the base and each lobe has two dorsal and three apical spines. *Echinogammarus ischnus* exhibits strong sexual dimorphism, with males having larger bodies and proportions than females. Male 2nd antennae are very setose and are almost the same length as the 1st antennae, while female 2nd antennae are visibly shorter than the 1st (Kohn and Waterstraat 1990; Witt et al. 1997).

Size: Adult length ranges from 6 to 15 mm for males and 5 to 13 mm for females (Kohn and Waterstraat 1990; Witt et al. 1997; Nalepa et al. 2001).

Native Range: Ponto-Caspian region, in both the Black Sea drainage and the Caspian Sea drainages (Cristescu et al. 2004).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences: *Echinogammarus ischnus* was first reported in 1994 from the Detroit River (Witt et al. 1997). Archived specimens show that it could have been present in western Lake Erie as early as 1993 (Van Overdijk et al. 2003). By 1996, *E. ischnus* was widely distributed from southern Lake Huron downstream to the mouth of the Niagara River of Lake Ontario (Dermott et al. 1998). It was reported from Lake Michigan in 1998 (Nalepa et al. 2001) and Lake Superior in 2001 (Grigorovich et al. 2003). It is now widespread in southern Lake Huron, the St. Clair River, the Lake Ontario watershed and the upper St. Lawrence River (Dermott et al. 1998; Palmer and Ricciardi 2004). This species expanded its range outside the Great Lake drainages into the Mississippi River drainage. It has been found in the lower half of the Ohio River since 2004 and in the upper portion in 2005 as well as in the upper Mississippi River from the confluence of the Ohio River upriver into Minnesota (Grigorovich et al. 2008).

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Echinogammarus ischnus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations?
IL	1999	2020	11	Apple-Plum; Chicago; Copperas-Duck; Des Plaines; Lake Michigan; Little Calumet-Galien; Lower Illinois-Lake Chautauqua; Lower Ohio-Bay; Pike-Root; The Sny; Upper Illinois
IN	1999	2016	3	Lake Michigan; Little Calumet-Galien; Lower Ohio
IA	2004	2006	4	Apple-Plum; Coon-Yellow; Copperas-Duck; Grant-Little Maquoketa
KY	2004	2006	6	Blue-Sinking; Lower Ohio-Bay; Lower Ohio-Little Pigeon; Middle Ohio-Laughery; Ohio Brush-Whiteoak; Silver-Little Kentucky
MI	1995	2023	15	Au Gres-Rifle; Carp-Pine; Detroit; Huron; Kawkawlin-Pine; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Lone Lake-Ocqueoc; Manistee; Muskegon; Ottawa-Stony; Pigeon-Wiscoggin; Saginaw
MN	2004	2020	4	Lake Superior; Rush-Vermillion; St. Louis; Twin Cities
NY	2007	2016	5	Chaumont-Perch; Lake Erie; Lake Ontario; Niagara River; Seneca
OH	1996	2023	5	Cuyahoga; Lake Erie; Lower Maumee; Ohio Brush-Whiteoak; Upper Ohio
PA	2007	2012	1	Lake Erie
WV	2005	2005	1	Upper Ohio
WI	2004	2017	10	Buffalo-Whitewater; Coon-Yellow; Door-Kewaunee; Duck-Pensaukee; Lake Michigan; Lake Superior; Lower Fox; Milwaukee; Peshtigo; Rush-Vermillion

Table last updated 12/8/2025

† Populations may not be currently present.

Ecology: *Echinogammarus ischnus* is a euryhaline species that is most common in large northern rivers of the Black and Caspian Sea drainages (Kohn and Waterstraat 1990; Cristescu et al. 2004). *Echinogammarus ischnus* is known to live and reproduce in waters ranging in salinity from 0 to 18 psu (Mordukhai-Boltovskoi 1964; Dediu 1980). It has been recorded at depths ranging from 0 to 300 m (typically <50 m) on mud, silt, sand, rock, *Dreissena* mussels and under wrack (Kohn and Waterstraat 1990; Witt et al. 1997; Nalepa et al. 2001; Grigorovich et al. 2003; Kley and Maier 2005). Its occurrence and density in the upper St. Lawrence River is positively correlated with current velocity and the availability of gravel-sized sediment (Palmer & Ricciardi 2004). It can tolerate highly eutrophic conditions and temperatures up to an absolute maximum of 33–35°C, with a thermal optimum around 27°C (Kohn and Waterstraat 1990; Wijnhoven et al. 2003; Romanenko et al. 2020). Because *E. ischnus* is a species from the Ponto-Caspian region that has high environmental variability, it and other invaders may benefit from future warming from climate change in the Great Lakes region (Casties et al. 2018).

Echinogammarus ischnus feeds on deposits, including those associated with zebra mussels, and preys on other macroinvertebrates such as chironomids, other amphipods, or its own species (Krisp and Maier 2005; Limen et al. 2005). In general, *E. ischnus* is more carnivorous than amphipods such as *Gammarus fasciatus* and *Hyalella azteca* (Limen et al. 2005). It is consumed by fish, including *Acipenser fulvescens* (Lake Sturgeon), *Perca flavescens* (Yellow Perch), and *Neogobius melanostomus* (Round Goby) (González and Burkhart 2004; Bailey 2015; Bruestle et al. 2019).

Means of Introduction: *Echinogammarus ischnus* was almost certainly introduced in ballast water (Witt et al. 1997). Its subsequent introduction to a port in Lake Superior (Grigorovich et al. 2003) was likely caused by a separate introduction in ballast water, either from a saltwater vessel or a domestic freighter from another freshwater port.

Status: This species has expanded its range outside the Great Lake drainages into the Mississippi River drainage.

Great Lakes:
Established where recorded in Lakes Erie, Huron, Ontario, and Michigan. It is assumed to not yet be established in Lake Superior because only two populations have been found to date (Grigorovich et al. 2003; Trebitz et al. 2010).

Impact of Introduction: A) **Realised:** *E. ischnus* is now among the most abundant non-*Dreissena* invertebrates in benthic communities in the Lake Ontario, L. Michigan, and L. Erie watersheds, where it has displaced native *Gmelinoides fasciatus* from many areas (Dermott et al. 1998; Stewart et al. 1998a, 1998b; Nalepa et al. 2001; Ratti and Barton 2003; van Overdijk et al. 2003; Haynes et al. 2005). Habitat heterogeneity in the St. Lawrence River may be promoting the co-existence of both species by allowing them to segregate along physicochemical gradients (Palmer & Ricciardi 2004). It is possible that *E. ischnus* has benefited from a co-evolved relationship with dreissenid mussels (Ricciardi & MacIsaac 2000). Structural complexity of *Dreissena* mussel substrate in combination with available nutrition from mussel biodeposits may have given *E. ischnus* a competitive advantage, stimulating its population expansion in the lower Great Lakes (van Overdijk et al. 2003). However, native amphipods consume dreissenid mussel pseudefeces more than the invader does at some sites, thus mussel habitat structure alone could enhance *E. ischnus* dominance over native species in the Great Lakes (Limen et al. 2005). *E. ischnus* in the St. Lawrence may be more susceptible to predation amongst dreissenid mussels than *G. fasciatus*, and predation may allow for coexistence of introduced and native species (Palmer and Ricciardi 2005). In the Great Lakes, however, *E. ischnus* is found more frequently amongst introduced mussels where native amphipods obtain refuge from predators amongst macrophytes and Cladophora-encrusted habitats (van Overdijk et al. 2003; González and Burkart 2004).

B) **Potential:** This species has also been introduced to Western Europe and the Baltic Sea (Cristescu et al. 2004). In parts of Germany and Poland, it has reduced or replaced native gammarids (such as *G. fossarum*, *G. roeseli* and *G. pulex*) (Jazdzewski et al. 2004; Kinzler and Maier 2006). In streams in Central Europe, introduced *E. ischnus* and *Dikerogammarus villosus* appear to have contributed to declines in some native macroinvertebrates through predation (Krisp and Maier 2005). It is possible that with increasing colonization of *D. bugensis* in deeper waters of the Great Lakes, *E. ischnus* may follow (Nalepa et al. 2001). Canal systems promote the dispersal of this species throughout Europe and may aid its further dispersal in North America (Witt et al. 1997).

Remarks: Although there is great genetic variation amongst native populations in the native Ponto-Caspian region, one mitochondrial genotype of this species from the Black Sea has been responsible for invasions from the Rhine River to North America. (Cristescu et al. 2004).

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Great Lakes Waterlife

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